OSAMA HAMADA

AI Engineer | Machine Learning | Deep Learning | Agentic Systems | Cloud Deployment

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PROFESSIONAL SUMMARY

AI Solutions Developer with hands-on expertise in Machine Learning, Deep Learning, Agentic Systems, and Computer Vision — building end-to-end intelligent systems from research to production. Proven industry experience architecting autonomous multi-agent platforms at Nafez AI and SEO intelligence pipelines at Chapters Digital Solutions. Competitive track record (Top 10% in international MTC-AIC3 competition), developing StarGAN v2 vision pipelines, and production RAG. Proficient in FastAPI, Docker, TiDB, MongoDB, SSE streaming, and cloud deployment.

EXPERIENCE

- **AI Engineer** | **Nafez AI** | Jul 2026 – Aug 2026 | Cairo, Egypt (Remote)

  Agentic Multi-Agent Systems: Architected an autonomous multi-agent system enabling dynamic reasoning, task decomposition, agent orchestration, and automated tool execution for a real-time e-commerce shopping assistant.

  LLM & Intent Detection: Engineered bilingual (Arabic/English) NLP pipelines for complex query intent understanding, parameter extraction, and intelligent request routing.

  Semantic Search Engine: Built semantic search pipelines using dense embeddings, cosine similarity, multi-attribute filtering, and re-ranking for natural-language product discovery.

  Real-Time Data Engineering: Engineered high-throughput concurrent FastAPI pipelines processing tens of thousands of live products across Amazon, Noon, Jumia, and B.TECH into TiDB.

  AI Microservices & Production: Developed standalone Python AI microservices with async workflows, SSE streaming for real-time low latency, and Docker production deployment.

- **AI Engineer** | **Chapters Digital Solutions** | Sep 2026 (1 mo) | Cairo, Egypt (Remote)

  Multi-Agent SEO Architecture: Architected an autonomous multi-agent system to automate SEO analysis, audit workflows, and actionable technical recommendations.

  Google Search Console Integration: Built automated pipelines interfacing with Google Search Console APIs to fetch, monitor, and analyze live indexing, search queries, impressions, CTR, and keyword rankings.

  Automated Report Optimization: Implemented LLM-driven agents to process API performance data, detect organic ranking anomalies, and autonomously edit and generate client-ready SEO audit reports.

- **AI and Vision Systems Lead** | **MTC-AIC3 International Competition** | Jun 2025 – Jul 2025 | Remote

  Led technical team to architect hierarchical neural architectures for EEG (SSVEP/MI), achieving 0.67039 accuracy and Rank 23 / 215 teams (Top 10%).

TECHNICAL SKILLS

- **Languages & Frameworks:** Python, C++, PyTorch, TensorFlow, FastAPI, Flask
- **Machine Learning & DL:** Hugging Face (Transformers, Hub, Pipelines, Fine-tuning), CNN, RNN, GAN (StarGAN v2), Transfer Learning, Model Ensembling, Feature Engineering
- **Agentic AI & LLMs:** Multi-Agent Architectures, LangGraph, LLM Workflows, Intent Detection, RAG (ChromaDB), Tool Calling, Prompt Engineering, Ollama, LangChain
- **Computer Vision:** OpenCV, MediaPipe, YOLO, Object Detection, Object Tracking, Image Segmentation, Face Alignment
- **Data & DevOps:** TiDB, MongoDB, Vector Databases, Docker, Microservices, Real-Time Data Pipelines, SSE, Git, Railway

PROJECTS

Neurina – Agentic AI Face Style Transfer System v2 | GitHub: github.com/osamahmada2024/Neurina

Multi-agent system for facial style transfer using StarGAN v2. Built 8 LangGraph agents orchestrating autonomous conversation-to-image pipeline with SSE streaming. Integrated StarGAN v2, CodeFormer + Real-ESRGAN face restoration, OpenCV preprocessing, ChromaDB RAG, and 3 Docker microservices (FastAPI + MongoDB) with Cloudinary.

mini_rag — Document Ingestion Pipeline for RAG Systems v1 | GitHub: github.com/osamahmada2024/mini_rag

FastAPI document ingestion service with LangChain integration. Implements file validation, intelligent text chunking via RecursiveCharacterTextSplitter, async MongoDB persistence (Motor), and clean Dockerized MVC architecture.

Movie Recommendation System v1 | GitHub: github.com/osamahmada2024/Movie-Recommender

Real-time recommendation engine for 10,000+ movie titles with sub-50ms retrieval using Sentence-BERT embeddings. Built with Flask, JavaScript, and React UI. Upgrading to v2 multi-modal API with hybrid similarity search.

DiagnosisXAi – Intelligent System for Medical Diagnosis v1 | GitHub: github.com/osamahmada2024/DiagnosisXAi

Diagnostic support system trained on 1.2M patient interactions (DDXPlus). Achieved Top-5 Accuracy of 94% through hyperparameter tuning and model ensembling, with SHAP values for XAI transparency. Deployed live on Railway.

EDUCATION

Bachelor of Computer Science, Faculty of Computers and Informatics, Tanta University (Expected 08/2027 | Tanta, Egypt)

CERTIFICATIONS

Develop Generative AI Applications: Get Started	**IBM**
AWS Academy Graduate – Cloud Foundations	**AWS**

LANGUAGES

Arabic: Native | English: Very Good